Simulation for Beginners - Projects - Example Processes

Example Process 1: Making Yogurt

Initial Scenario: Fermentation of Milk

- **Description:** Milk is first pasteurized and then combined with starter cultures (e.g., lactic acid bacteria) to initiate the fermentation process. This step involves adding the starter culture to a fermenter and maintaining controlled fermentation conditions at a specific temperature and duration.
- **Input:** Pasteurized milk and starter culture.
- **Output:** Fermented milk (yogurt base mass).

Example Process 2: Production of a Biologically Active Ingredient

Initial Scenario: Cultivation of Microorganisms

- **Description:** In a bioreactor, microorganisms (e.g., bacteria, yeasts, or cell lines) are cultivated to produce a biologically active ingredient. The process involves adding a sterile culture medium to the bioreactor to enable microbial growth and active ingredient production.
- **Input:** Sterile culture medium and starter culture (e.g., bacteria).
- **Output:** Culture medium containing the produced active ingredient (e.g., antibodies, proteins).

Example Process 3: Making Chocolate

Initial Scenario: Cocoa Bean Processing

- **Description:** The process begins with the cleaning and roasting of cocoa beans. The beans are then crushed and ground to obtain cocoa mass. Sugar and, if applicable, milk are added to the cocoa mass. Finally, the mixture is refined until a smooth chocolate mass is achieved, which serves as the foundation for various chocolate products.
- **Input:** Cocoa beans, sugar, milk.
- **Output:** Smooth chocolate mass.

Example Process 4: Patient Processing in the Emergency Room

Initial Scenario: Initial Triage of an Emergency Patient

- **Description:** A patient enters the emergency room (or is transported by ambulance). The patient undergoes an initial assessment by a triage team (e.g., a nurse) to determine the severity of the emergency. Based on the assessment, the patient is either treated immediately or directed to the waiting area according to priority.
- **Input:** Patient with acute complaints.
- **Output:** Patient is assigned a priority level (e.g., placed in a treatment category).